

Aishwarya Mohanraj

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PROFESSIONAL SUMMARY

Health Informatics graduate student at Indiana University Indianapolis with a clinical background in dentistry and hands-on experience in biomedical research, healthcare analytics, and clinical data management. Skilled in Python, SQL, R, FHIR interoperability, EHR systems, and machine learning, with a strong interest in developing data-driven solutions to improve healthcare delivery and patient outcomes.

EDUCATION

Indiana University Indianapolis

Master of Science in Health Informatics.

Indianapolis, IN

Expected August 2027

SRM KDC

Bachelor of Dental Surgery.

Chennai, India

December 2020

EXPERIENCE

Research Data Analyst, Healthcare Analytics and Biomedical Research

Sep 2025 – Present

Indiana University Indianapolis

Indianapolis, IN

- Analyze MOX sensor datasets using Python (NumPy, Pandas) for VOC profile analysis in non-invasive disease detection.
- Develop preprocessing and feature extraction pipelines to convert sensor signals into structured datasets for machine learning.
- Perform statistical analysis and feature engineering to identify VOC patterns supporting non-invasive disease detection research.

Teaching Assistant, Health Information Standards and Terminologies

Aug 2026 – Present

Indiana University Indianapolis

Indianapolis, IN

- Guide students on HL7, FHIR, SNOMED CT, LOINC, and ICD standards; conduct tutoring sessions and assist with grading.

Public Health Data Analyst

Sep 2021 – Jul 2025

Abrasives and Chemicals

Tiruchirappalli, India

- Analyzed public health datasets using Python, SQL, and Excel to identify population health trends and risk factors.

EHR Analyst

Mar 2021 – Aug 2021

Mahathma Gandhi Memorial Hospital

Tiruchirappalli, Tamil Nadu, India

- Maintained EHR records by documenting patient histories, diagnoses, and treatment plans to support continuity of care.

Clinical Research Coordinator

Sep 2019 – Jan 2021

SRM Kattankulathur Dental College

Chennai, India

- Coordinated clinical research, managed patient data collection, and maintained clinical records to support research studies.

PROJECTS

FHIR Healthcare ETL Pipeline

Jan 2026 – May 2026

- Designed an ETL pipeline to extract patient data from OpenEMR via FHIR APIs and transform it into standardized interoperable formats.
- Standardized terminology using SNOMED CT and generated HL7 v2 messages to support interoperability and data exchange.
- Developed documentation and workflow visualizations using Python, FHIR APIs, OpenEMR, SNOMED CT, and HL7 v2.

Autoimmune Biomarker Analysis

Jan 2026 – May 2026

- Analyzed 12,500+ patient records using R and statistical modeling to identify autoimmune disease predictors.

Childhood Vaccination Data Analytics

Aug 2025 – Dec 2025

- Analyzed SDOH and vaccination data across 92 Indiana counties using Python, SQL, and machine learning.

MedMind: AI-Enabled Digital Mental Health Platform

Aug 2025 – Dec 2025

- Designed an AI-enabled platform with wearable integration, HIPAA-compliant architecture, and a \$453K budget.

SKILLS

Health Informatics: EHR/EMR Systems, FHIR, HL7 v2, SNOMED CT, LOINC, ICD, SDOH, Public Health Data

Programming Languages: Python, SQL, R, SAS

Data Analytics and Machine Learning: Pandas, NumPy, Statistical Analysis, PCA, Feature Engineering, Data Preprocessing, Chemometrics, Random Forest, Logistic Regression

Data Visualization: Power BI, Tableau, Matplotlib, Seaborn

Tools: Jupyter Notebook, Git/GitHub, MS Excel, Visual Studio Code, MS Project

CERTIFICATIONS

- SNOMED CT Foundation Course, SNOMED International (Issued Aug 2026)
- Basic Life Support, American Heart Association